1.12 Homework: Review of Sequences and Series — Markscheme

1. [Maximum mark: 7]

   $7, 11, 15, 19, \ldots$, so $u_1 = 7$.

   (a) $d = 4$ **A1**

   (b) $u_n = 7 + (n-1)4$ **M1**

   $u_n = 4n + 3$ **A1**

   Accept the unsimplified form $7 + 4(n-1)$ *for both marks.*

   (c) $u_{25} = 4(25) + 3 = 103$ **A1**

   (d) $4n + 3 = 143$ **M1**

   $4n = 140$ **(A1)**

   $n = 35$: the 35th term **A1**

   M1 for setting the expression for u_n *equal to* 143. *FT from (b).*

2. [Maximum mark: 7]

   (a) $u_1 + 2d = 14$ and $u_1 + 7d = 39$ **M1**

   $5d = 25$, so $d = 5$ **A1**

   $u_1 = 14 - 2(5) = 4$ **A1**

   M1 for both equations, or for $\frac{39-14}{8-3}$. *Accept a GDC simultaneous-equation solve.*

   (b) $u_{20} = 4 + 19(5) = 99$ **(M1)A1**

   (c) $S_{20} = \dfrac{20}{2}(4 + 99)$ **M1**

   $= 1030$ **A1**

   Accept $S_{20} = \frac{20}{2}(2(4) + 19(5))$. *FT from (a) and (b).*

3. [Maximum mark: 4]

$u_1 = 5$, $d = 4$, last term 201.

(a) $5 + (n-1)4 = 201$ **M1**

$n = 50$ **A1**

The common error is $\frac{201-5}{4} = 49$*, forgetting to add* 1*: M1A0.*

(b) $S_{50} = \frac{50}{2}(5 + 201)$ **M1**

$= 5150$ **A1**

FT from the candidate's n *in (a).*

4. [Maximum mark: 7]

$u_1 = 20$, $d = 2$, 25 rows.

(a) $u_{25} = 20 + 24(2) = 68$ seats **(M1)A1**

(b) $S_{25} = \frac{25}{2}(20 + 68)$ **M1**

$= 1100$ seats **A1**

FT from (a).

(c) $\frac{n}{2}(2(20) + (n-1)2) \geq 600$, i.e. $n(n+19) \geq 600$ **M1**

$S_{16} = 560$ and $S_{17} = 612$ **(A1)**

17 rows **A1**

M1 for setting up the sum of the first n *rows* ≥ 600 *(or* $= 600$*). The GDC table or the quadratic* $n^2 + 19n - 600 = 0$*, giving* $n = 16.77\ldots$*, then rounding* up *to* 17*. The answer* 16 *(rounding down) earns M1A1A0:* 16 *rows hold only* 560 *seats.*

5. [Maximum mark: 7]

   $3, 6, 12, 24, \ldots$, so $u_1 = 3$.

   (a) $r = 2$ **A1**

   (b) $u_n = 3 \times 2^{n-1}$ **A1**

   6^{n-1} *(multiplying* 3×2 *first) earns A0.*

   (c) $u_{10} = 3 \times 2^9 = 1536$ **(M1)A1**

   (d) $3 \times 2^{n-1} > 10\,000$ **M1**

   $u_{12} = 6144, \quad u_{13} = 12\,288$ **(A1)**

   The 13th term, 12 288 **A1**

   M1 for the inequality, a GDC table, or listing terms. Final A1 needs both the value and the term number.

6. [Maximum mark: 6]

   (a) $u_1 r = 12$ and $u_1 r^4 = 324$ **M1**

   $r^3 = \dfrac{324}{12} = 27$, so $r = 3$ **A1**

   $u_1 = \dfrac{12}{3} = 4$ **A1**

   (b) $u_7 = 4 \times 3^6 = 2916$ **A1**

   (c) $S_6 = \dfrac{4\,(3^6 - 1)}{3 - 1}$ **M1**

   $= \dfrac{4(728)}{2} = 1456$ **A1**

   FT from (a). $u_6 = 972$ *(a term, not a sum) earns no marks.*

7. [Maximum mark: 5]

(a) $r = 1.1$ **A1**

$r = 0.1$ or $r = 10$ earns A0.

(b) $u_8 = 5 \times 1.1^7$ **(M1)**

$= 9.7435\ldots = 9.74$ km **A1**

(c) $S_8 = \dfrac{5\,(1.1^8 - 1)}{1.1 - 1}$ **(M1)**

$= 57.179\ldots = 57.2$ km **A1**

An exponent of 8 in (b), giving 10.7, earns M1A0. FT from (a).

8. [Maximum mark: 8]

***GDC:** the Finance Solver may be used with $N = 4n$, $I(\%) = 4$, $PV = -5000$, $PMT = 0$, $P/Y = C/Y = 4$.*

(a) $FV = 5000\left(1 + \dfrac{4}{100 \times 4}\right)^{4\times 6} = 5000(1.01)^{24}$ **M1**

$= €6348.67$ **A1**

Annual compounding in (a) (6326.60) earns M0A0.

(b) $6348.67 - 5000 = €1348.67$ **A1**

FT from (a).

(c) $5000(1.04)^6 = €6326.60$ **(M1)A1**

(d) $5000(1.01)^{4n} > 8000$ **M1**

After 11 years: €7746.59; after 12 years: €8061.13 **(A1)**

12 years **A1**

M1 for the inequality, a GDC table, or the Finance Solver with $FV = 8000$ ($N = 47.2$ quarters). 11.8 years, unrounded, earns M1A1A0; the question asks for complete years.